Share in a ratio

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: Share £420 in the ratio 3:4. Copy or complete the first step.

2. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

3. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Share £420 in the ratio 3:4. Copy or complete the first step.
   Answer £180 and £240.

2. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
   Answer 52 km.

3. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
   Answer 24.5 cm.

4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.
   Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

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Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Share £420 in the ratio 3:4.

2. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

3. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

4. Solve this independently and show every stage: Share £420 in the ratio 3:4.

5. Use a second method or representation for: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

6. Check this result using an inverse, estimate or known fact: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

7. Prove or explain your reasoning: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

Teacher answers and checking prompts

1. Share £420 in the ratio 3:4.
   Answer £180 and £240.

2. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
   Answer 52 km.

3. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
   Answer 24.5 cm.

4. Solve this independently and show every stage: Share £420 in the ratio 3:4.
   Answer £180 and £240.

5. Use a second method or representation for: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
   Answer Expected result: 52 km. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
   Answer Expected result: 24.5 cm. A valid check must be shown.

7. Prove or explain your reasoning: A recipe for 6 uses 450 g flour. How much flour is needed for 14?
   Answer 1,050 g. The explanation must show why the method works.

8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.
   Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ____________

1. Prove or explain your reasoning: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

2. Prove the answer to this question, rather than only calculating it: Share £420 in the ratio 3:4.

3. Solve this in two different ways and compare them: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

4. Work backwards from the answer to reconstruct the question: 24.5 cm.

5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.

6. Create a new example that tests the same idea as: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A recipe for 6 uses 450 g flour. How much flour is needed for 14?
   Answer 1,050 g. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Share £420 in the ratio 3:4.
   Answer Expected result: £180 and £240. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
   Answer Expected result: 52 km. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 24.5 cm.
   Answer One valid reconstruction is: A length of 7 cm is enlarged by scale factor 3.5. What is the new length? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.
   Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

6. Create a new example that tests the same idea as: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.